Scientific Analysis of Moral Judgments using Tobit and Cluster-Aware Non-Parametric Robustness Models

Automated Research Pipeline

marzo 27, 2026

1 Dataset and Sample Description

The empirical foundation of this project rests on two primary experimental datasets: FLORIDA and BUC. The sample consists of students from the Bucaramanga Campus. These datasets capture incentivized moral judgments from distinct socio-economic contexts. Participants were presented with standardized negotiation scenarios where a negotiator's decision resulted in varying degrees of payoff for themselves, their own group, and a victim group (either ingroup or outgroup).

2 Datacard and Variable Definitions

The following table defines the primary symbols and variables used in the mathematical specifications and hypothesis tests.

Table 1: Symbols and Variable Dictionary.

Symbol	Definition
y_{ij}	Observed moral judgment of scenario j by participant i.
y^*_{ij}	Latent moral preference score (unbounded).
β_1	Regression coefficient representing the marginal effect of the predictor.
$\beta_{0.5}$	Median-regression coefficient from the non-parametric censored robustness model.
$Q_{0.5}(y^*_{ij} \mid \mathbf{x}_{ij})$	Conditional median of the latent bounded outcome given the predictors.
IRI_i	Empathy score (Average composite of the Interpersonal Reactivity Index).
OutgroupPerp	Binary indicator: 1 if the scenario perpetrator is an outgroup member.
SameGroupHarm	Binary indicator: 1 if the harm is inflicted on the perpetrator's own group.
ICC	Intraclass Correlation: ratio of between-cluster variance to total variance.

3 Hypotheses to Test

- **H1 (Empathy):** Higher empathy predicts lower moral-judgment scores for harmful decisions.
- **H2a (Betrayal):** Same-faculty harm receives lower moral-judgment scores than cross-faculty harm.
- **H2b (Outgroup):** Outgroup perpetrators receive lower moral-judgment scores than ingroup perpetrators.
- **H3 (Moderation):** The empathy effect is stronger in outgroup cases.

4 Mathematical Approach and Theoretical Foundations

The analysis employs two complementary estimators for bounded moral judgments. The primary specification is a Two-Limit Tobit model, which is theoretically appropriate for dependent variables that are strictly bounded within a known interval. In this context, moral judgments (y_{ij}) are observed on a scale from -9 to 9. The Tobit model assumes the existence of a latent, unobserved preference index (y^*_{ij}) that follows a linear relationship:

$$y^*_{ij} = \mathbf{x}'_{ij}\beta + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

The actual observed judgment y_{ij} relates to this latent variable via the censoring transformation:

$$y_{ij} = \max(-9, \min(9, y^*_{ij}))$$

This approach prevents the 'ceiling' and 'floor' effects from biasing the linear coefficients, as would occur in standard OLS regression.

Because the Tobit model relies on a Gaussian latent-error assumption, the pipeline also fits a distribution-robust censored median specification implemented as interval-censored quantile regression ($p = 0.5$). This complementary estimator targets the conditional median of the latent bounded outcome and is less sensitive to heavy tails and non-normal disturbances:

$$Q_{0.5}(y^*_{ij} \mid \mathbf{x}_{ij}) = \mathbf{x}'_{ij}\beta_{0.5}$$

The non-parametric branch preserves the censoring structure while relaxing the parametric normality assumption. The Tobit and robustness results should therefore be interpreted jointly: Tobit provides the clustered parametric benchmark with participant-clustered standard errors by id, while the non-parametric branch first establishes a converged full-sample censored median fit and then, in the default pipeline, adds participant-level cluster bootstrap inference that resamples ids with replacement and retains all repeated observations from each sampled participant. In both branches, id is used only to account for within-participant dependence and is not a substantive explanatory variable.

5 Analysis of Sample Size Impact

The statistical validity of our inferences depends on the trade-off between False Positives (Type I error, α) and False Negatives (Type II error, β). Given the clustered nature of our data, sample size impact is not merely a count of observations, but a function of cluster correlation.

5.0.1 Step-by-Step Sensitivity Analysis

To determine the impact of our sample size on our ability to detect effects, we follow these steps:

1. Calculate the Design Effect (Deff): As multiple judgments are nested within individuals, we adjust for the Intraclass Correlation (ICC).

$$Deff = 1 + (m - 1) \times ICC$$

where m is the average number of scenarios per participant.

2. Determine the Effective Sample Size (ESS): The ESS represents the number of independent observations that would provide the same statistical power as our clustered sample.

$$ESS = \frac{n_{total}}{Deff}$$

3. Inference Impact: A higher ICC reduces the ESS, thereby increasing the Standard Error of our Tobit coefficients. If the ESS is low, our models become 'conservative', increasing the risk of Type II errors (failing to support a hypothesis). By using clustered robust standard errors for Tobit and participant-level cluster bootstrap inference for the non-parametric robustness model after each converged full-sample fit, we ensure that both sets of p-values acknowledge this reduced information density, protecting the integrity of our Type I error threshold ($\alpha = 0.05$).

Based on the current run, the observed Intraclass Correlation (ICC) is 0.717, with an Effective Sample Size (ESS) of 266.9.

6 Descriptive Statistics

The empathy profile of the sample is visualized in Figure 1, showing the average scores across the four IRI latent variables.

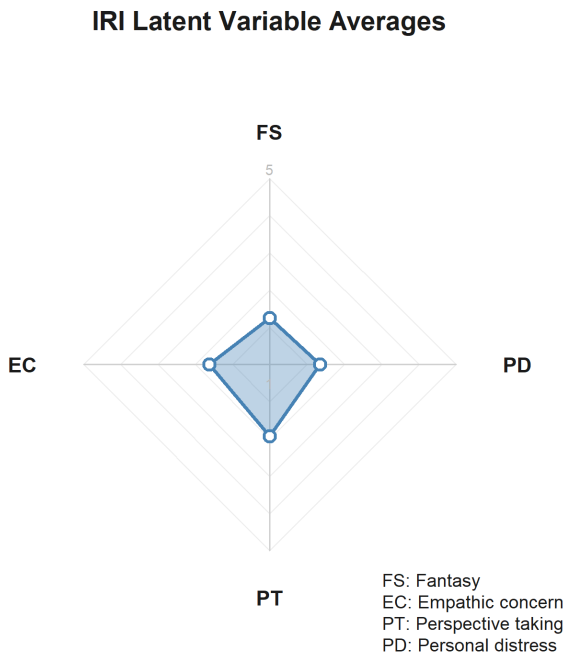

Figure 1: IRI Latent Variable Averages (Radar Plot profile).

Judgment severity distributions segmented by experimental conditions are presented in Figure 2.

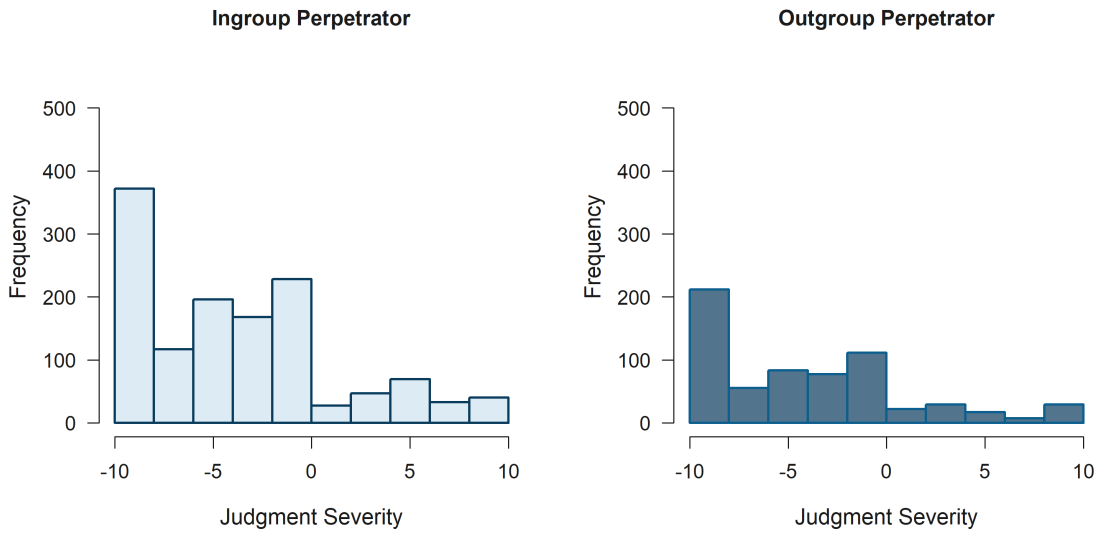

Figure 2: Judgment Severity Distribution (Ingroup vs. Outgroup).

7 Bi-variate Statistics

The correlation matrix between the psychometric subscales and the mean moral judgment is presented below.

Table 2: Correlation Matrix: IRI Subscales and Moral Judgment.

Fantasy	Empathic Concern	Perspective Taking	Personal Distress	Total IRI	Mean Judgment
1.000	0.500	0.554	0.509	0.830	-0.030
0.500	1.000	0.424	0.649	0.818	-0.116
0.554	0.424	1.000	0.301	0.730	0.074
0.509	0.649	0.301	1.000	0.762	-0.010
0.830	0.818	0.730	0.762	1.000	-0.029
-0.030	-0.116	0.074	-0.010	-0.029	1.000

Visual representations of these relationships are provided in Figure 3.

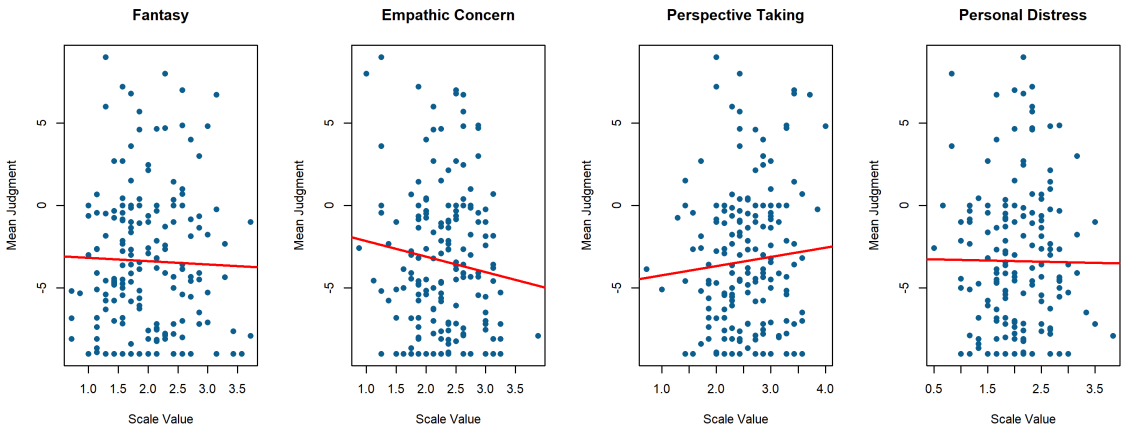

Figure 3: Bivariate Scatters: IRI Scales vs. Mean Judgment.

8 Estimator Fit Summary

The following table consolidates the fit-status information for the primary Tobit models and the non-parametric robustness branch. In the default pipeline, participant-level cluster bootstrap launches immediately after a converged full-sample non-parametric fit is available; if bootstrap is disabled manually or too few bootstrap refits converge, that status is shown explicitly.

Table 3: Estimator fit summary across Tobit and cluster-aware non-parametric specifications.

Model	Approach	Status	Converged	Iterations	BootstrapReplicates	BootstrapSuccessful	Observations	Participants	ClusterUnit
H1_A_Total_CLAD	CLAD	not_converged	FALSE	2000	10	NA	1939	199	id
H1_B_Constructs_CLAD	CLAD	completed	TRUE	93	10	4	1939	199	id
H2a_A_Total_CLAD	CLAD	completed	TRUE	43	10	10	1262	199	id
H2a_B_Constructs_CLAD	CLAD	completed	TRUE	27	10	4	1262	199	id
H2b_A_Total_CLAD	CLAD	not_converged	FALSE	2000	10	NA	1939	199	id
H2b_B_Constructs_CLAD	CLAD	completed	TRUE	93	10	3	1939	199	id
H3_A_Total_CLAD	CLAD	not_converged	FALSE	2000	10	NA	1939	199	id
H3_B_Constructs_CLAD	CLAD	completed	TRUE	93	10	3	1939	199	id
H1_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H1_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H2a_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1262	199	id
H2a_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1262	199	id
H2b_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H2b_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H3_A_Total_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id
H3_B_Constructs_Tobit	Tobit	completed	NA	3	NA	NA	1939	199	id

8.1 Hypothesis Significance Summary

The following concise table lists only hypothesis-relevant predictors that reached at least $p < 0.10$, using conventional symbols to indicate strength of evidence. If the non-parametric bootstrap is disabled, too sparse, or the censored median fit does not converge, the non-parametric column reports that status instead of inferential symbols. Dynamic figures are generated only for predictors that appear in this table with at least one significance symbol.

Table 4: Hypothesis-level significance summary across Tobit and cluster-aware non-parametric models.

Hypothesis	Tobit significant predictors	Non-parametric significant predictors
Higher empathy predicts lower moral-judgment scores for harmful decisions.	Empathy: Perspective taking+	None
Same-faculty harm receives lower moral-judgment scores than cross-faculty harm.	None	None
Outgroup perpetrators receive lower moral-judgment scores than ingroup perpetrators.	None	None
The empathy effect is stronger in outgroup cases than in ingroup cases.	Personal distress x outgroup perpetrator**; Empathic concern x outgroup perpetrator**	Personal distress x outgroup perpetrator**; Empathic concern x outgroup perpetrator**

8.2 Significance-Driven Figures

Only hypothesis-relevant predictors that reach $p < 0.10$ or better are visualized automatically. These dynamic figures use the saved Tobit and clustered non-parametric outputs, and participant id remains only an inference-level clustering unit rather than a substantive explanatory variable.

H1: Empathy effect Empathy: Perspective taking is statistically significant in the Tobit model (+, $p = 0.064$). Figure 4 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted latent judgment.

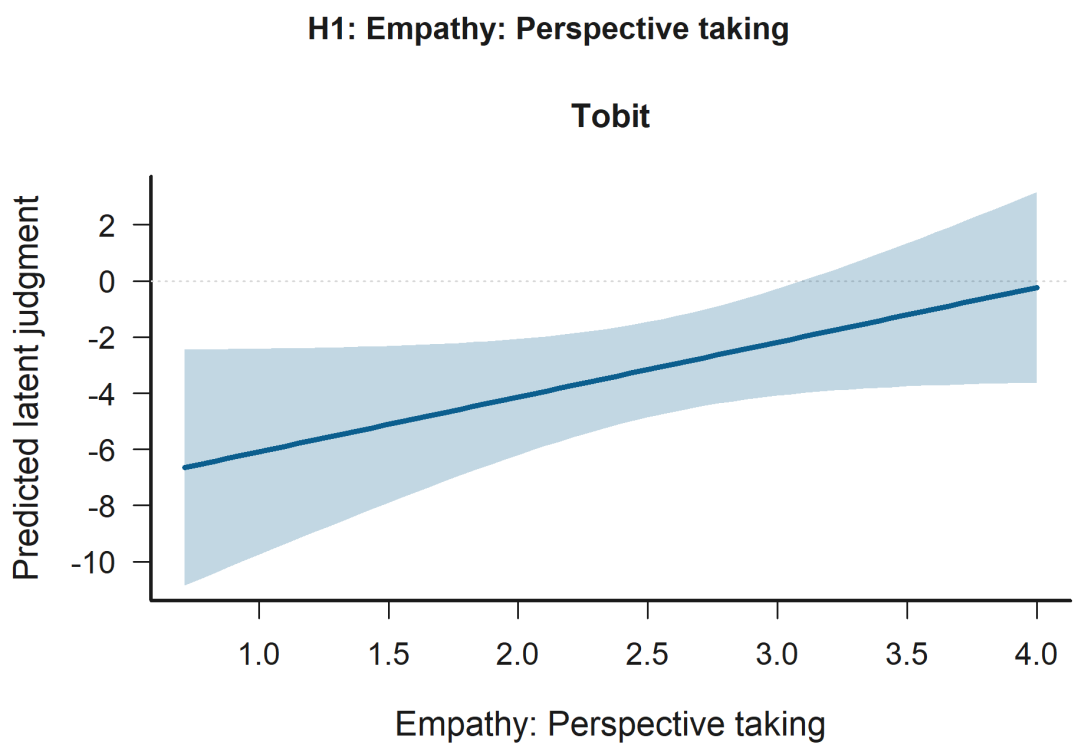

Figure 4: Effect Plot for Empathy: Perspective taking in H1: Empathy effect. Support comes from the Tobit model (+, $p = 0.064$). The panels show predicted latent judgments with 95% confidence intervals.

H3: Moderation Personal distress x outgroup perpetrator is statistically significant in the clustered non-parametric model (**, $p < 0.001$) and the Tobit model (**, $p < 0.001$). Figure 5 shows that both estimator panels indicate that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is Outgroup perpetrator.

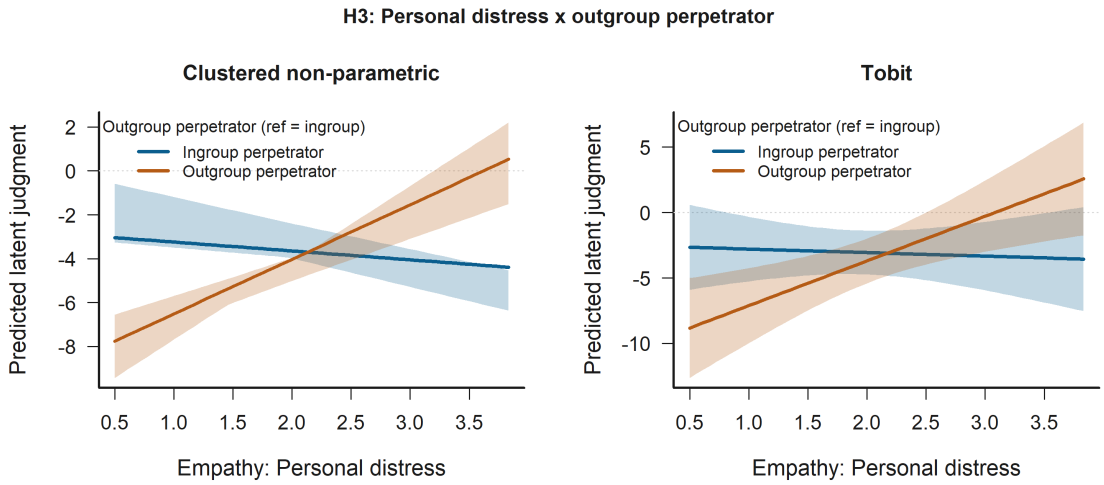

Figure 5: Interaction Plot for Personal distress x outgroup perpetrator in H3: Moderation. Support comes from the clustered non-parametric model (**, $p < 0.001$) and the Tobit model (**, $p < 0.001$). The panels show predicted latent judgments with 95% confidence intervals.

Empathic concern x outgroup perpetrator is statistically significant in the clustered non-parametric model (**, $p < 0.001$) and the Tobit model (**, $p = 0.010$). Figure 6 shows that both estimator panels indicate that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is Outgroup perpetrator.

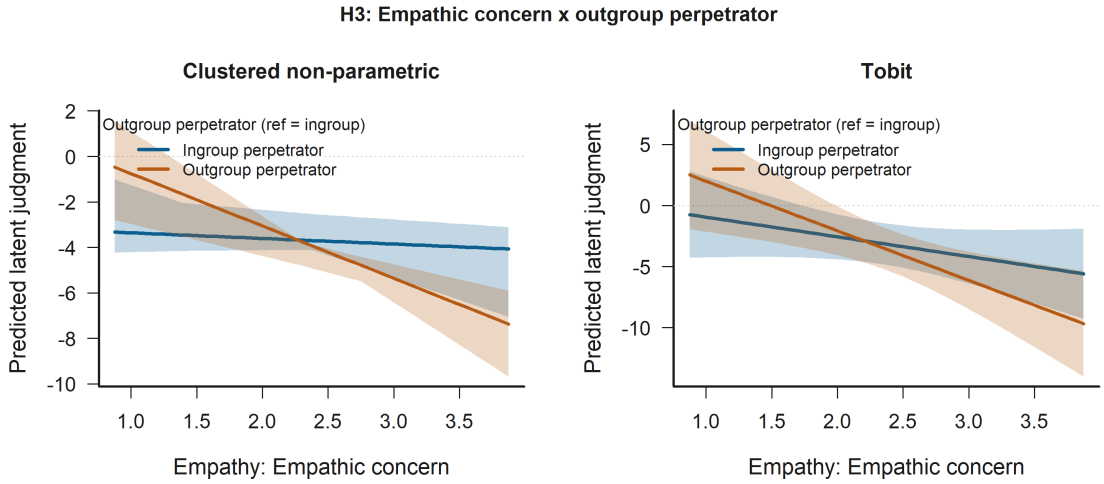

Figure 6: Interaction Plot for Empathic concern x outgroup perpetrator in H3: Moderation. Support comes from the clustered non-parametric model (**, $p < 0.001$) and the Tobit model (**, $p = 0.010$). The panels show predicted latent judgments with 95% confidence intervals.

8.3 Integrated Hypothesis Conclusions

The following summary restates each original hypothesis and indicates whether the available Tobit estimates and the cluster-aware non-parametric models support it in the current data. Non-

parametric conclusions are drawn when the participant-level bootstrap inference is available and are otherwise labeled explicitly.

- H1. Original hypothesis: Higher empathy predicts lower moral-judgment scores for harmful decisions. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; Empathy composite (average) is negative but not statistically significant ($p = 0.517$). Model B does not support the hypothesis; none of the empathy subscale main effects are statistically significant, and the closest signal is Empathy: Perspective taking with a positive association ($p = 0.064$). Additional statistically significant signals include Personal distress x outgroup perpetrator with a positive association ($p < 0.001$) and Empathic concern x outgroup perpetrator with a negative association ($p = 0.010$). Non-parametric conclusion: the available models do not support the hypothesis. Only non-parametric specifications with available cluster-bootstrap inference are interpreted here. Model A is not interpreted because the non-parametric optimization did not converge after 2000 iterations. Model B does not support the hypothesis; none of the empathy subscale main effects are statistically significant, and the closest signal is Empathy: Perspective taking with a positive association ($p = 0.347$). Additional statistically significant signals include Empathic concern x outgroup perpetrator with a negative association ($p < 0.001$) and Personal distress x outgroup perpetrator with a positive association ($p < 0.001$).

- H2a. Original hypothesis: Same-faculty harm receives lower moral-judgment scores than cross-faculty harm. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; Negotiator and victim share faculty is negative but not statistically significant ($p = 0.751$). Model B does not support the hypothesis; Negotiator and victim share faculty is negative but not statistically significant ($p = 0.670$). Additional statistically significant signals include Observer role (ref = victim) with a positive association ($p = 0.013$) and Empathy: Empathic concern with a negative association ($p = 0.014$). Non-parametric conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; Negotiator and victim share faculty is positive but not statistically significant ($p = 0.535$). Model B does not support the hypothesis; Negotiator and victim share faculty is positive but not statistically significant ($p = 0.477$). Additional statistically significant signals include Age with a positive association ($p = 0.008$) and Observer role (ref = victim) with a positive association ($p = 0.025$).

- H2b. Original hypothesis: Outgroup perpetrators receive lower moral-judgment scores than ingroup perpetrators. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; Outgroup perpetrator (ref = ingroup) is negative but not statistically significant ($p = 0.930$). Model B does not support the hypothesis; Outgroup perpetrator (ref = ingroup) is positive but not statistically significant ($p = 0.740$). Additional statistically significant signals include Personal distress x outgroup perpetrator with a positive association ($p < 0.001$) and Empathic concern x outgroup perpetrator with a negative association ($p = 0.010$). Non-parametric conclusion: the available models do not support the hypothesis. Only non-parametric specifications with available cluster-bootstrap inference are interpreted here. Model A is not interpreted because the non-parametric optimization did not converge after 2000 iterations. Model B does not support the hypothesis; Outgroup perpetrator (ref = ingroup) is negative but not statistically significant ($p = 0.949$). Additional statistically significant signals include Empathic concern x outgroup perpetrator with a negative association ($p < 0.001$) and Personal distress x outgroup perpetrator with a positive association ($p < 0.001$).

- H3. Original hypothesis: The empathy effect is stronger in outgroup cases than in ingroup cases. Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A does not support the hypothesis; Empathy x outgroup perpetrator is negative but not statistically significant ($p = 0.928$). Model B supports the hypothesis through Empathic concern x outgroup perpetrator with a negative association ($p = 0.010$). Additional statistically significant signals include Observer role (ref = victim) with a positive association ($p = 0.016$) and Age with a positive association ($p = 0.034$). Non-parametric conclusion: the available models support the hypothesis. Only non-parametric specifications with available cluster-bootstrap inference are interpreted here. Model A is not interpreted because the non-parametric optimization did not converge after 2000 iterations. Model B supports the hypothesis through Empathic concern x outgroup perpetrator with a negative association ($p < 0.001$). Additional statistically significant signals include Observer role (ref = victim) with a positive association ($p = 0.002$) and Age with a positive association ($p = 0.015$).

9 Hypothesis Validation and Results

Detailed coefficient tables for each Tobit model, coupled with non-parametric robustness outputs, natural language interpretive narratives, and estimator-specific diagnostics, are provided below. In the default pipeline, converged non-parametric fits immediately attempt participant-level cluster-bootstrap inference; the report labels deferred and sparse-bootstrap cases explicitly when full inference is not available.

9.1 H1: Empathy Effect

This section evaluates the primary effect of empathy on moral judgments.

9.1.1 Model A: Composite Empathy

Table 5: H1 Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.574	4.801	0.005
Empathy composite (average)	-0.693	1.070	0.517
Outgroup perpetrator (ref = ingroup)	-0.176	2.018	0.930
Control label hidden (ref = ingroup)	0.266	1.908	0.889
Victim outgroup (ref = ingroup)	-0.029	0.499	0.953
Observer role (ref = victim)	0.929	0.429	0.030
Engineering participant (ref = humanities)	1.545	0.877	0.078
Man (ref = woman)	0.346	0.896	0.700
Age	0.412	0.194	0.034
Socioeconomic status	0.265	0.393	0.501
Negotiator 2 (ref = negotiator 1)	0.620	0.261	0.017
Empathy x outgroup perpetrator	-0.078	0.860	0.928
Empathy x control label hidden	0.081	0.837	0.923
Scale variance (log)	1.893	0.060	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.574 (p=0.005). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.693. The p-value of 0.517 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.176. The p-value of 0.930 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.266. The p-value of 0.889 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.029. The p-value of 0.953 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.929. The p-value of 0.030 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.545. The p-value of 0.078 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing

whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.346. The p-value of 0.700 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.412. The p-value of 0.034 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.265. The p-value of 0.501 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_outgroup (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of -0.078. The p-value of 0.928 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 0.081. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.893, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.953, p = 1.832e-24). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 6: H1 Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-15.134	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	-0.773	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Outgroup perpetrator (ref = ingroup)	-2.095	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Control label hidden (ref = ingroup)	-0.527	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Victim outgroup (ref = ingroup)	0.236	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.901	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.085	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.208	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.604	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	-0.006	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.192	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy x outgroup perpetrator	0.902	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy x control label hidden	0.363	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.134 (p=NA). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.773. The p-value of NA the p-value cannot be determined. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.095. The p-value of NA the p-value cannot be determined. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -0.527. The p-value of NA the p-value cannot be determined.

The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.236. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.901. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.085. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.208. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.604. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.006. The p-value of NA the p-value cannot be determined. The term iri_total:perp_outgroup (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.902. The p-value of NA the p-value cannot be determined. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 0.363. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.1.2 Model B: Separated Empathy Constructs

Table 7: H1 Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.716	4.938	0.005
Empathy: Fantasy scale	-0.463	0.900	0.606
Empathy: Empathic concern	-1.613	1.084	0.137
Empathy: Perspective taking	1.952	1.054	0.064
Empathy: Personal distress	-0.269	0.978	0.783
Outgroup perpetrator (ref = ingroup)	0.717	2.161	0.740
Control label hidden (ref = ingroup)	0.227	2.173	0.917
Victim outgroup (ref = ingroup)	-0.307	0.478	0.521
Observer role (ref = victim)	1.016	0.421	0.016
Engineering participant (ref = humanities)	1.458	0.887	0.100
Man (ref = woman)	0.192	0.942	0.838
Age	0.383	0.193	0.048
Socioeconomic status	0.099	0.389	0.799
Negotiator 2 (ref = negotiator 1)	0.598	0.259	0.021
Fantasy x outgroup perpetrator	-0.822	0.696	0.237
Empathic concern x outgroup perpetrator	-2.444	0.944	0.010
Perspective taking x outgroup perpetrator	-0.548	0.872	0.530
Personal distress x outgroup perpetrator	3.688	0.868	0.000
Fantasy x control label hidden	-0.567	0.773	0.464
Empathic concern x control label hidden	-0.248	1.070	0.817
Perspective taking x control label hidden	-0.089	0.912	0.923
Personal distress x control label hidden	1.027	0.898	0.253
Scale variance (log)	1.870	0.057	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.716 (p=0.005). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.463. The p-value of 0.606 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.613. The p-value of 0.137 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an es-

timated β coefficient of 1.952. The p-value of 0.064 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.269. The p-value of 0.783 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of 0.717. The p-value of 0.740 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.227. The p-value of 0.917 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.307. The p-value of 0.521 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.016. The p-value of 0.016 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.458. The p-value of 0.100 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.192. The p-value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.383. The p-value of 0.048 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.099. The p-value of 0.799 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.822. The p-value of 0.237 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_outgroup (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -2.444. The p-value of 0.010 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt:perp_outgroup (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.548. The p-value of 0.530 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_outgroup (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 3.688. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.567. The p-value of 0.464 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator)

has an estimated β coefficient of -0.248. The p-value of 0.817 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of -0.089. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 1.027. The p-value of 0.253 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.870, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.964, p = 1.750e-21). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 8: H1 Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-15.603	5.656	0.006
Empathy: Fantasy scale	-1.028	1.718	0.550
Empathy: Empathic concern	-0.246	1.118	0.826
Empathy: Perspective taking	0.884	0.940	0.347
Empathy: Personal distress	-0.405	0.896	0.651
Outgroup perpetrator (ref = ingroup)	-0.171	1.967	0.931
Control label hidden (ref = ingroup)	0.297	1.733	0.864
Victim outgroup (ref = ingroup)	0.144	0.209	0.491
Observer role (ref = victim)	0.847	0.285	0.003
Engineering participant (ref = humanities)	0.239	1.080	0.825
Man (ref = woman)	0.334	0.496	0.500
Age	0.591	0.220	0.007
Socioeconomic status	-0.062	0.480	0.898
Negotiator 2 (ref = negotiator 1)	0.190	0.190	0.317
Fantasy x outgroup perpetrator	0.675	1.467	0.645
Empathic concern x outgroup perpetrator	-2.055	0.554	0.000
Perspective taking x outgroup perpetrator	-1.015	1.187	0.392
Personal distress x outgroup perpetrator	2.891	0.787	0.000
Fantasy x control label hidden	0.138	1.313	0.916
Empathic concern x control label hidden	-0.706	0.407	0.083
Perspective taking x control label hidden	-0.110	0.983	0.910
Personal distress x control label hidden	0.823	0.983	0.403

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.603 (p=0.006). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.028. The p-value of 0.550 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.246. The p-value of 0.826 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.884. The p-value of 0.347 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress

and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.405. The p-value of 0.651 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.171. The p-value of 0.931 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.297. The p-value of 0.864 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.144. The p-value of 0.491 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.847. The p-value of 0.003 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.239. The p-value of 0.825 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.334. The p-value of 0.500 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.591. The p-value of 0.007 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.062. The p-value of 0.898 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of 0.675. The p-value of 0.645 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_outgroup (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -2.055. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt:perp_outgroup (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -1.015. The p-value of 0.392 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_outgroup (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 2.891. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of 0.138. The p-value of 0.916 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of -0.706. The p-value of 0.083 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control

(representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of -0.110. The p-value of 0.910 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 0.823. The p-value of 0.403 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 10 ids with replacement, retaining all repeated observations from each sampled participant; 4 bootstrap refits converged successfully. The full-sample optimization status is converged after 93 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

9.2 H2a: Ingroup Betrayal

This section tests whether harm directed at the perpetrator's own group is judged differently.

9.2.1 Model A: Composite Empathy Control

Table 9: H2a Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-14.098	4.503	0.002
Negotiator and victim share faculty	-0.142	0.446	0.751
Empathy composite (average)	-0.764	1.039	0.462
Outgroup perpetrator (ref = ingroup)	-0.418	0.442	0.343
Victim outgroup (ref = ingroup)	0.055	0.606	0.928
Observer role (ref = victim)	1.153	0.497	0.020
Engineering participant (ref = humanities)	1.075	0.924	0.245
Man (ref = woman)	0.404	0.940	0.668
Age	0.438	0.188	0.020
Socioeconomic status	0.466	0.392	0.235
Negotiator 2 (ref = negotiator 1)	0.444	0.340	0.192
Scale variance (log)	1.895	0.063	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated

at -14.098 (p=0.002). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β coefficient of -0.142. The p-value of 0.751 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.764. The p-value of 0.462 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.418. The p-value of 0.343 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.055. The p-value of 0.928 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.153. The p-value of 0.020 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.075. The p-value of 0.245 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.404. The p-value of 0.668 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.438. The p-value of 0.020 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.466. The p-value of 0.235 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.895, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.947, p = 5.693e-21). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 10: H2a Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-16.132	4.806	0.001
Negotiator and victim share faculty	0.296	0.477	0.535
Empathy composite (average)	-0.434	0.817	0.595
Outgroup perpetrator (ref = ingroup)	-0.110	0.440	0.803
Victim outgroup (ref = ingroup)	0.372	0.275	0.176
Observer role (ref = victim)	0.859	0.392	0.029
Engineering participant (ref = humanities)	-0.251	1.127	0.823
Man (ref = woman)	0.483	0.504	0.338
Age	0.579	0.218	0.008
Socioeconomic status	0.199	0.299	0.506
Negotiator 2 (ref = negotiator 1)	0.373	0.273	0.171

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -16.132 (p=0.001). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β coefficient of 0.296. The p-value of 0.535 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.434. The p-value of 0.595 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.110. The p-value of 0.803 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.372. The p-value of 0.176 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.859. The p-value of 0.029 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.251. The p-value of 0.823 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.483. The p-value of 0.338 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.579. The p-value of 0.008 indicates statistically significant, meaning

the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.199. The p-value of 0.506 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 10 ids with replacement, retaining all repeated observations from each sampled participant; 10 bootstrap refits converged successfully. The full-sample optimization status is converged after 43 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

9.2.2 Model B: Separated Construct Controls

Table 11: H2a Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.233	4.876	0.007
Negotiator and victim share faculty	-0.193	0.453	0.670
Empathy: Fantasy scale	-0.933	0.868	0.283
Empathy: Empathic concern	-2.914	1.184	0.014
Empathy: Perspective taking	1.596	0.974	0.101
Empathy: Personal distress	1.633	0.974	0.094
Outgroup perpetrator (ref = ingroup)	-0.420	0.437	0.336
Victim outgroup (ref = ingroup)	-0.158	0.590	0.789
Observer role (ref = victim)	1.200	0.482	0.013
Engineering participant (ref = humanities)	1.042	0.940	0.268
Man (ref = woman)	0.230	0.991	0.816
Age	0.400	0.194	0.039
Socioeconomic status	0.316	0.407	0.437
Negotiator 2 (ref = negotiator 1)	0.272	0.333	0.414
Scale variance (log)	1.878	0.060	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.233 (p=0.007). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β coefficient of -0.193. The p-value of 0.670 indicates the effect is not statistically significant, meaning

we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.933. The p-value of 0.283 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.914. The p-value of 0.014 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.596. The p-value of 0.101 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.633. The p-value of 0.094 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.420. The p-value of 0.336 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.158. The p-value of 0.789 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.200. The p-value of 0.013 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.042. The p-value of 0.268 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.230. The p-value of 0.816 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.400. The p-value of 0.039 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.316. The p-value of 0.437 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.878, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.960, p = 3.724e-18). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 12: H2a Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-16.182	6.376	0.011
Negotiator and victim share faculty	0.133	0.188	0.477
Empathy: Fantasy scale	-0.599	1.056	0.570
Empathy: Empathic concern	-1.178	0.893	0.187
Empathy: Perspective taking	0.245	0.805	0.760
Empathy: Personal distress	0.852	0.637	0.181
Outgroup perpetrator (ref = ingroup)	-0.114	0.491	0.816
Victim outgroup (ref = ingroup)	0.134	0.312	0.668
Observer role (ref = victim)	1.113	0.498	0.025
Engineering participant (ref = humanities)	-0.039	0.985	0.969
Man (ref = woman)	0.439	0.506	0.386
Age	0.605	0.261	0.020
Socioeconomic status	0.236	0.328	0.471
Negotiator 2 (ref = negotiator 1)	0.258	0.454	0.570

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -16.182 (p=0.011). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β coefficient of 0.133. The p-value of 0.477 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.599. The p-value of 0.570 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.178. The p-value of 0.187 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.245. The p-value of 0.760 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.852. The p-value of 0.181 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.114. The p-value of 0.816 indicates the effect is not statistically significant, meaning we do not have enough evidence

to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.134. The p-value of 0.668 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.113. The p-value of 0.025 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.039. The p-value of 0.969 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.439. The p-value of 0.386 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.605. The p-value of 0.020 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.236. The p-value of 0.471 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 10 ids with replacement, retaining all repeated observations from each sampled participant; 4 bootstrap refits converged successfully. The full-sample optimization status is converged after 27 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

9.3 H2b: Outgroup Derogation

This section examines the difference in judgment when the perpetrator is an outgroup member compared to an ingroup member.

9.3.1 Model A: Composite Empathy Control

Table 13: H2b Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.574	4.801	0.005
Outgroup perpetrator (ref = ingroup)	-0.176	2.018	0.930
Empathy composite (average)	-0.693	1.070	0.517
Control label hidden (ref = ingroup)	0.266	1.908	0.889
Victim outgroup (ref = ingroup)	-0.029	0.499	0.953
Observer role (ref = victim)	0.929	0.429	0.030
Engineering participant (ref = humanities)	1.545	0.877	0.078
Man (ref = woman)	0.346	0.896	0.700
Age	0.412	0.194	0.034
Socioeconomic status	0.265	0.393	0.501
Negotiator 2 (ref = negotiator 1)	0.620	0.261	0.017
Empathy x outgroup perpetrator	-0.078	0.860	0.928
Empathy x control label hidden	0.081	0.837	0.923
Scale variance (log)	1.893	0.060	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.574 (p=0.005). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.176. The p-value of 0.930 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.693. The p-value of 0.517 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.266. The p-value of 0.889 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.029. The p-value of 0.953 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.929. The p-value of 0.030 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.545. The p-value of 0.078 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing

whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.346. The p-value of 0.700 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.412. The p-value of 0.034 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.265. The p-value of 0.501 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_total (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of -0.078. The p-value of 0.928 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 0.081. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.893, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.953, p = 1.832e-24). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 14: H2b Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-15.134	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Outgroup perpetrator (ref = ingroup)	-2.095	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	-0.773	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Control label hidden (ref = ingroup)	-0.527	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Victim outgroup (ref = ingroup)	0.236	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.901	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.085	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.208	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.604	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	-0.006	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.192	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy x outgroup perpetrator	0.902	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy x control label hidden	0.363	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.134 (p=NA). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.095. The p-value of NA the p-value cannot be determined. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.773. The p-value of NA the p-value cannot be determined. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -0.527. The p-value of NA the p-value cannot be determined.

The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.236. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.901. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.085. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.208. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.604. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.006. The p-value of NA the p-value cannot be determined. The term perp_outgroup:iri_total (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.902. The p-value of NA the p-value cannot be determined. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 0.363. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.3.2 Model B: Separated Construct Controls

Table 15: H2b Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.716	4.938	0.005
Outgroup perpetrator (ref = ingroup)	0.717	2.161	0.740
Empathy: Fantasy scale	-0.463	0.900	0.606
Empathy: Empathic concern	-1.613	1.084	0.137
Empathy: Perspective taking	1.952	1.054	0.064
Empathy: Personal distress	-0.269	0.978	0.783
Control label hidden (ref = ingroup)	0.227	2.173	0.917
Victim outgroup (ref = ingroup)	-0.307	0.478	0.521
Observer role (ref = victim)	1.016	0.421	0.016
Engineering participant (ref = humanities)	1.458	0.887	0.100
Man (ref = woman)	0.192	0.942	0.838
Age	0.383	0.193	0.048
Socioeconomic status	0.099	0.389	0.799
Negotiator 2 (ref = negotiator 1)	0.598	0.259	0.021
Fantasy x outgroup perpetrator	-0.822	0.696	0.237
Empathic concern x outgroup perpetrator	-2.444	0.944	0.010
Perspective taking x outgroup perpetrator	-0.548	0.872	0.530
Personal distress x outgroup perpetrator	3.688	0.868	0.000
Fantasy x control label hidden	-0.567	0.773	0.464
Empathic concern x control label hidden	-0.248	1.070	0.817
Perspective taking x control label hidden	-0.089	0.912	0.923
Personal distress x control label hidden	1.027	0.898	0.253
Scale variance (log)	1.870	0.057	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.716 (p=0.005). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of 0.717. The p-value of 0.740 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.463. The p-value of 0.606 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient

of -1.613. The p-value of 0.137 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.952. The p-value of 0.064 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.269. The p-value of 0.783 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.227. The p-value of 0.917 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.307. The p-value of 0.521 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.016. The p-value of 0.016 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.458. The p-value of 0.100 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.192. The p-value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.383. The p-value of 0.048 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.099. The p-value of 0.799 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_fs (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.822. The p-value of 0.237 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -2.444. The p-value of 0.010 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.548. The p-value of 0.530 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 3.688. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.567. The p-value of 0.464 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator)

has an estimated β coefficient of -0.248. The p-value of 0.817 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of -0.089. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 1.027. The p-value of 0.253 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.870, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.964, p = 1.750e-21). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 16: H2b Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-15.603	5.169	0.003
Outgroup perpetrator (ref = ingroup)	-0.171	2.676	0.949
Empathy: Fantasy scale	-1.028	1.044	0.325
Empathy: Empathic concern	-0.246	0.575	0.668
Empathy: Perspective taking	0.884	0.828	0.286
Empathy: Personal distress	-0.405	0.925	0.662
Control label hidden (ref = ingroup)	0.297	1.792	0.869
Victim outgroup (ref = ingroup)	0.144	0.370	0.698
Observer role (ref = victim)	0.847	0.230	0.000
Engineering participant (ref = humanities)	0.239	1.756	0.892
Man (ref = woman)	0.334	0.814	0.681
Age	0.591	0.244	0.015
Socioeconomic status	-0.062	0.358	0.863
Negotiator 2 (ref = negotiator 1)	0.190	0.137	0.164
Fantasy x outgroup perpetrator	0.675	0.867	0.436
Empathic concern x outgroup perpetrator	-2.055	0.420	0.000
Perspective taking x outgroup perpetrator	-1.015	0.284	0.000
Personal distress x outgroup perpetrator	2.891	0.650	0.000
Fantasy x control label hidden	0.138	0.444	0.757
Empathic concern x control label hidden	-0.706	0.434	0.104
Perspective taking x control label hidden	-0.110	0.541	0.838
Personal distress x control label hidden	0.823	0.748	0.272

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.603 (p=0.003). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.171. The p-value of 0.949 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.028. The p-value of 0.325 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.246. The p-value of 0.668 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological

point of view of others (Perspective Taking)) has an estimated β coefficient of 0.884. The p-value of 0.286 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.405. The p-value of 0.662 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.297. The p-value of 0.869 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.144. The p-value of 0.698 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.847. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.239. The p-value of 0.892 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.334. The p-value of 0.681 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.591. The p-value of 0.015 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.062. The p-value of 0.863 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_fs (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of 0.675. The p-value of 0.436 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -2.055. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -1.015. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 2.891. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of 0.138. The p-value of 0.757 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of -0.706. The p-value of 0.104 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction

effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of -0.110. The p-value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 0.823. The p-value of 0.272 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 10 ids with replacement, retaining all repeated observations from each sampled participant; 3 bootstrap refits converged successfully. The full-sample optimization status is converged after 93 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

9.4 H3: Interaction Moderation

This section tests the interaction and moderation between absolute empathy dimensions and the outgroup status of the perpetrator.

9.4.1 Model A: Composite Empathy Interaction

Table 17: H3 Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.574	4.801	0.005
Empathy composite (average)	-0.693	1.070	0.517
Outgroup perpetrator (ref = ingroup)	-0.176	2.018	0.930
Control label hidden (ref = ingroup)	0.266	1.908	0.889
Victim outgroup (ref = ingroup)	-0.029	0.499	0.953
Observer role (ref = victim)	0.929	0.429	0.030
Engineering participant (ref = humanities)	1.545	0.877	0.078
Man (ref = woman)	0.346	0.896	0.700
Age	0.412	0.194	0.034
Socioeconomic status	0.265	0.393	0.501
Negotiator 2 (ref = negotiator 1)	0.620	0.261	0.017
Empathy x outgroup perpetrator	-0.078	0.860	0.928
Empathy x control label hidden	0.081	0.837	0.923
Scale variance (log)	1.893	0.060	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.574 (p=0.005). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.693. The p-value of 0.517 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.176. The p-value of 0.930 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.266. The p-value of 0.889 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.029. The p-value of 0.953 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.929. The p-value of 0.030 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.545. The p-value of 0.078 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.346. The p-value of 0.700 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.412. The p-value of 0.034 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.265. The p-value of 0.501 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_outgroup (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of -0.078. The p-value of 0.928 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 0.081. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.893, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.953, p = 1.832e-24). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 18: H3 Model A: Composite Empathy Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	inference
Intercept	-15.134	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy composite (average)	-0.773	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Outgroup perpetrator (ref = ingroup)	-2.095	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Control label hidden (ref = ingroup)	-0.527	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Victim outgroup (ref = ingroup)	0.236	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Observer role (ref = victim)	0.901	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Engineering participant (ref = humanities)	-0.085	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Man (ref = woman)	0.208	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Age	0.604	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Socioeconomic status	-0.006	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Negotiator 2 (ref = negotiator 1)	0.192	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy x outgroup perpetrator	0.902	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge
Empathy x control label hidden	0.363	Cluster bootstrap skipped because the full-sample non-parametric fit did not converge

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.134 (p=NA). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.773. The p-value of NA the p-value cannot be determined. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.095. The p-value of NA the p-value cannot be determined. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -0.527. The p-value of NA the p-value cannot be determined. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.236. The p-value of NA the p-value cannot be determined. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.901. The p-value of NA the p-value cannot be determined. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.085. The p-value of NA the p-value cannot be determined. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.208. The p-value of NA the p-value cannot be determined. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.604. The p-value of NA the p-value cannot be determined. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.006. The p-value of NA the p-value cannot be determined. The term iri_total:perp_outgroup (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.902. The p-value of NA the p-value cannot be determined. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 0.363. The p-value of NA the p-value cannot be determined.

Diagnostics

The non-parametric robustness model did not achieve convergence in the full sample after 2000 iterations, so participant-level cluster bootstrap inference was not attempted for this specification. The reported point estimates remain on the original judgement scale because the internal response shift of 10.0 units was back-transformed.

9.4.2 Model B: Separated Constructs Interaction

Table 19: H3 Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-13.716	4.938	0.005
Empathy: Fantasy scale	-0.463	0.900	0.606
Outgroup perpetrator (ref = ingroup)	0.717	2.161	0.740
Empathy: Empathic concern	-1.613	1.084	0.137
Empathy: Perspective taking	1.952	1.054	0.064
Empathy: Personal distress	-0.269	0.978	0.783
Control label hidden (ref = ingroup)	0.227	2.173	0.917
Victim outgroup (ref = ingroup)	-0.307	0.478	0.521
Observer role (ref = victim)	1.016	0.421	0.016
Engineering participant (ref = humanities)	1.458	0.887	0.100
Man (ref = woman)	0.192	0.942	0.838
Age	0.383	0.193	0.048
Socioeconomic status	0.099	0.389	0.799
Negotiator 2 (ref = negotiator 1)	0.598	0.259	0.021
Fantasy x outgroup perpetrator	-0.822	0.696	0.237
Empathic concern x outgroup perpetrator	-2.444	0.944	0.010
Perspective taking x outgroup perpetrator	-0.548	0.872	0.530
Personal distress x outgroup perpetrator	3.688	0.868	0.000
Fantasy x control label hidden	-0.567	0.773	0.464
Empathic concern x control label hidden	-0.248	1.070	0.817
Perspective taking x control label hidden	-0.089	0.912	0.923
Personal distress x control label hidden	1.027	0.898	0.253
Scale variance (log)	1.870	0.057	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -13.716 (p=0.005). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.463. The p-value of 0.606 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of 0.717. The p-value of 0.740 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient

of -1.613. The p-value of 0.137 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.952. The p-value of 0.064 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.269. The p-value of 0.783 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.227. The p-value of 0.917 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.307. The p-value of 0.521 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.016. The p-value of 0.016 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.458. The p-value of 0.100 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.192. The p-value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.383. The p-value of 0.048 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.099. The p-value of 0.799 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.822. The p-value of 0.237 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -2.444. The p-value of 0.010 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.548. The p-value of 0.530 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 3.688. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.567. The p-value of 0.464 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator)

has an estimated β coefficient of -0.248. The p-value of 0.817 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of -0.089. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 1.027. The p-value of 0.253 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.870, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.964, p = 1.750e-21). The primary Tobit model still uses participant-clustered standard errors by id, and the report pairs it with a cluster-bootstrap non-parametric censored robustness model.

Table 20: H3 Model B: Separated Constructs Regression Coefficients (cluster-aware non-parametric robustness).

label	estimate	std_error	p_value
Intercept	-15.603	6.432	0.015
Empathy: Fantasy scale	-1.028	1.510	0.496
Outgroup perpetrator (ref = ingroup)	-0.171	2.443	0.944
Empathy: Empathic concern	-0.246	1.166	0.833
Empathy: Perspective taking	0.884	0.828	0.286
Empathy: Personal distress	-0.405	0.500	0.418
Control label hidden (ref = ingroup)	0.297	2.072	0.886
Victim outgroup (ref = ingroup)	0.144	0.252	0.569
Observer role (ref = victim)	0.847	0.273	0.002
Engineering participant (ref = humanities)	0.239	1.357	0.861
Man (ref = woman)	0.334	0.304	0.271
Age	0.591	0.242	0.015
Socioeconomic status	-0.062	0.547	0.910
Negotiator 2 (ref = negotiator 1)	0.190	0.055	0.001
Fantasy x outgroup perpetrator	0.675	1.696	0.690
Empathic concern x outgroup perpetrator	-2.055	0.454	0.000
Perspective taking x outgroup perpetrator	-1.015	1.514	0.502
Personal distress x outgroup perpetrator	2.891	0.554	0.000
Fantasy x control label hidden	0.138	1.417	0.923
Empathic concern x control label hidden	-0.706	0.352	0.045
Perspective taking x control label hidden	-0.110	1.218	0.928
Personal distress x control label hidden	0.823	0.921	0.372

Non-parametric Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -15.603 (p=0.015). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.028. The p-value of 0.496 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.171. The p-value of 0.944 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.246. The p-value of 0.833 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological

point of view of others (Perspective Taking)) has an estimated β coefficient of 0.884. The p-value of 0.286 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.405. The p-value of 0.418 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of 0.297. The p-value of 0.886 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.144. The p-value of 0.569 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.847. The p-value of 0.002 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.239. The p-value of 0.861 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.334. The p-value of 0.271 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.591. The p-value of 0.015 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.062. The p-value of 0.910 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of 0.675. The p-value of 0.690 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -2.055. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -1.015. The p-value of 0.502 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 2.891. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of 0.138. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of -0.706. The p-value of 0.045 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt:perp_control (representing the combined interaction effect between Perspective

Taking and an unidentified perpetrator) has an estimated β coefficient of -0.110. The p-value of 0.928 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 0.823. The p-value of 0.372 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The non-parametric robustness model was estimated as an interval-censored median regression (p = 0.50), and participant id was used only as the clustering unit for inference rather than as a substantive predictor. Within-participant dependence is handled through a participant-level cluster bootstrap that resampled 10 ids with replacement, retaining all repeated observations from each sampled participant; 3 bootstrap refits converged successfully. The full-sample optimization status is converged after 93 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units, and the reported intercept is back-transformed to the original judgement scale. Reported standard errors come from that cluster bootstrap, percentile confidence intervals use the bootstrap distribution directly, and p-values are summarized from the bootstrap standard errors on a normal-approximation scale.

10 Discussion and Limitations

While the dual-estimator strategy strengthens the analysis, several limitations remain. First, both inference strategies assume independence between participants, which may still be violated by broader shared institutional or contextual shocks. Second, we assume a normal distribution for the latent errors ϵ_{ij} in the Tobit branch; departures from normality could affect the consistency of the maximum likelihood estimators. The non-parametric robustness branch reduces dependence on that assumption and adds participant-level cluster bootstrap inference after converged full-sample fits, but bootstrap summaries still carry Monte Carlo error and can be sensitive to convergence problems in difficult specifications. Finally, the use of unstandardized averages for empathy assumes a linear mapping between the psychometric scale and the latent moral preference across both estimators.

11 Conclusion

Based on the combined interval-censored Tobit estimations and the cluster-aware non-parametric robustness workflow, the structural effects of empathy, group dynamics, and betrayal have been documented alongside their theoretical assumptions above.